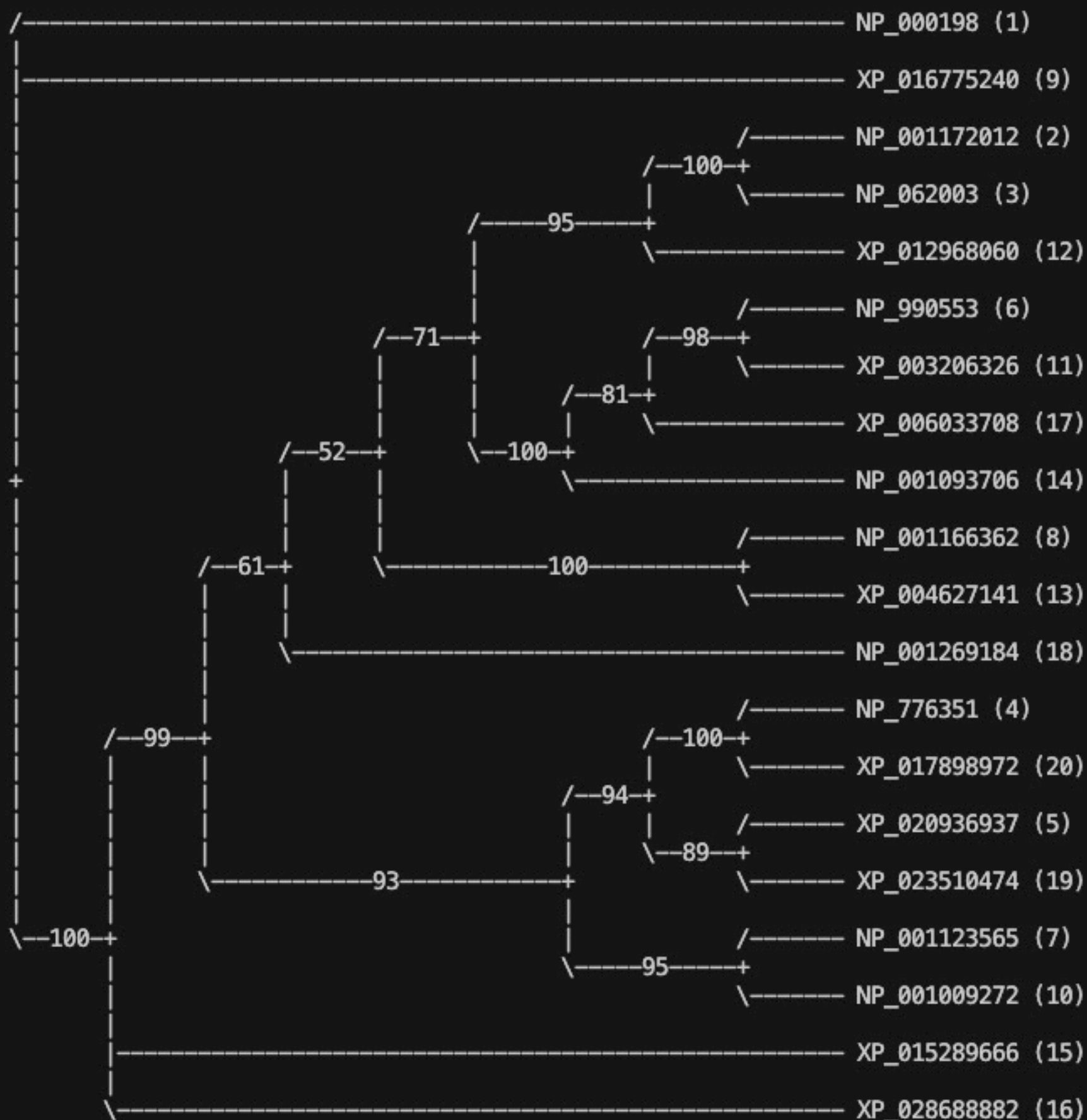


Clade credibility values:



```
1  # Name: Christopher A. Lee
2  # Date: 03/17/2026
3  # Prof: Dr. Callura
4  # Course: Bioinformatics - ENBC 455
5
6  # Libraries
7  from Bio import SeqIO
8
9  # Problem 2
10 for seq in SeqIO.parse("hw7_2.fasta", "fasta"):
11     print(seq.id)
12     print(seq.seq)
13
14 # Problem 3
15 with open("hw7_3.txt", "r") as f:
16     threshold = int(f.readline().strip())
17     count = 0
18
19     for seq in SeqIO.parse(f, "fastq"):
20         quality_scores = seq.letter_annotations['phred_quality']
21         avg_quality = sum(quality_scores) / len(quality_scores)
22         if avg_quality < threshold:
23             count += 1
24
25     print()
26     print(count, "\n")
27
28 # Problem 4
29 with open("hw7_4.txt", "r") as f:
30     first_line = f.readline().split()
31     q_threshold = int(first_line[0])
32     p_percentage = int(first_line[1])
33     count = 0
34
35     for seq in SeqIO.parse(f, "fastq"):
36         quality_scores = seq.letter_annotations["phred_quality"]
37         bases_above_q = sum(1 for score in quality_scores if score >= q_threshold)
38         current_p = (bases_above_q / len(quality_scores)) * 100
39
40         if current_p >= p_percentage:
41             count += 1
42
43     print(count)
44
```


Rosalind_0000
TGCTGACCGTTTACAGTCTGTTGACGAGGTCCCCTCTCCGAGGGACCGTGACCACATCCACCGATTTTAGGTTAAGGCGTACTCAACGTCCAAAATTGTACGACCATCAGATGGCACTGTTTGTGCGATTGAACCTACTGGGGTAACCCACGCGACTCCATA
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christopherlee@Christophers-Mac-mini-2 FNR455 - HW07 % ■

Rosalind_0089

GACTATTTTCTTAATATCCGGTTTCTCTGCTCTACGCATCTAATCGGTGCGTGCTGCTCGCTAGACGGGGCACGCTGACTGGACCCTTCCTCCAGAGGGGAACATAGCATTAGCGACCATGTCGTTCCAAACTGCTTACGCCTAACGATTAACAGCGATGTAC

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Rosalind_0091

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